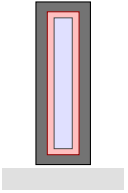
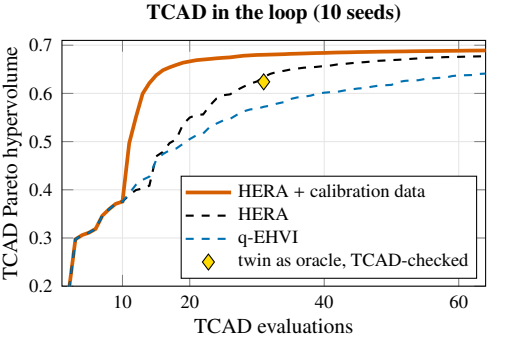
GAA FET
L_g 12–20 nm



6 variables,
5 constraints,
3 objectives

DEVSIM TCAD
96 + 32 calibration designs

Calibrated digital twin
held-out R^2: SS 0.93

Twin as oracle:
HERA, then TCAD check

TCAD in the loop:
HERA + calibration data